Lakehouse Architecture

A unified approach to data storage, processing, and analytics — built for the modern data stack.

DATA ENGINEERING

ARCHITECTURE OVERVIEW

What is a Lakehouse?

The Lakehouse is a modern data architecture paradigm that merges the best of two worlds: the vast, cost-effective storage of a **data lake** and the reliability, governance, and query performance of a **data warehouse**.

Data Lake

Stores raw, unstructured, and semi-structured data at massive scale. Flexible but hard to govern and query efficiently.

Data Warehouse

Delivers structured, governed, high-performance analytics — but at high cost and with limited support for unstructured data and ML workloads.

The Lakehouse brings ACID transactions, schema enforcement, and BI-grade performance directly onto open object storage — eliminating the need to maintain two separate systems.

Key Characteristics

A true Lakehouse platform is defined by three foundational properties that distinguish it from both legacy lakes and traditional warehouses.

ACID Transactions

Guarantees atomicity, consistency, isolation, and durability — even across concurrent reads and writes on large-scale datasets. No more partial updates or silent data corruption.

Schema Enforcement & Evolution

Prevents bad data from entering the system by validating against a defined schema at write time, while still allowing controlled schema changes as business needs evolve.

Query Performance

Leverages techniques such as data skipping, Z-ordering, caching, and columnar formats to deliver warehouse-class query speed directly on object storage.

Lake vs. Warehouse vs. Lakehouse

Understanding the trade-offs across architectures helps teams make informed decisions about where to invest.

Feature	Data Lake	Data Warehouse	Lakehouse
Storage cost	Low	High	Low
ACID transactions	No	Yes	Yes
Schema enforcement	Optional	Strict	Strict
Unstructured data	Yes	No	Yes
ML / AI workloads	Yes	Limited	Yes
BI & SQL analytics	Limited	Yes	Yes
Data versioning	No	No	Yes
Open format	Yes	No	Yes

The Lakehouse inherits strengths from both paradigms whilst eliminating most of the compromises — making it the dominant architecture choice for modern data platforms.

Benefits of the Lakehouse Model

Single Unified System

Consolidate your data estate onto one platform. Eliminate the costly and error-prone process of syncing data between a lake and a warehouse — one source of truth for all teams.

Lower Total Cost of Ownership

Object storage (e.g. S3, ADLS, GCS) is dramatically cheaper than proprietary warehouse storage. Pay for compute only when you need it, and scale independently of storage.

Multi-Workload Support

The same data layer supports BI dashboards, ad-hoc SQL, batch ETL pipelines, streaming ingestion, and ML model training — without duplication or data movement.

Medallion Architecture

The Medallion Architecture is the recommended data organisation pattern within a Lakehouse. It structures data across three progressive quality layers, each serving a distinct purpose in the data lifecycle.

1

🥉 Bronze

Raw ingestion layer. Data lands here exactly as received — unmodified, with full history preserved. Acts as the system of record for all source data.

2

🥈 Silver

Cleansed & conformed layer. Data is deduplicated, validated, type-cast, and joined across sources. This is the reliable domain model teams build on.

3

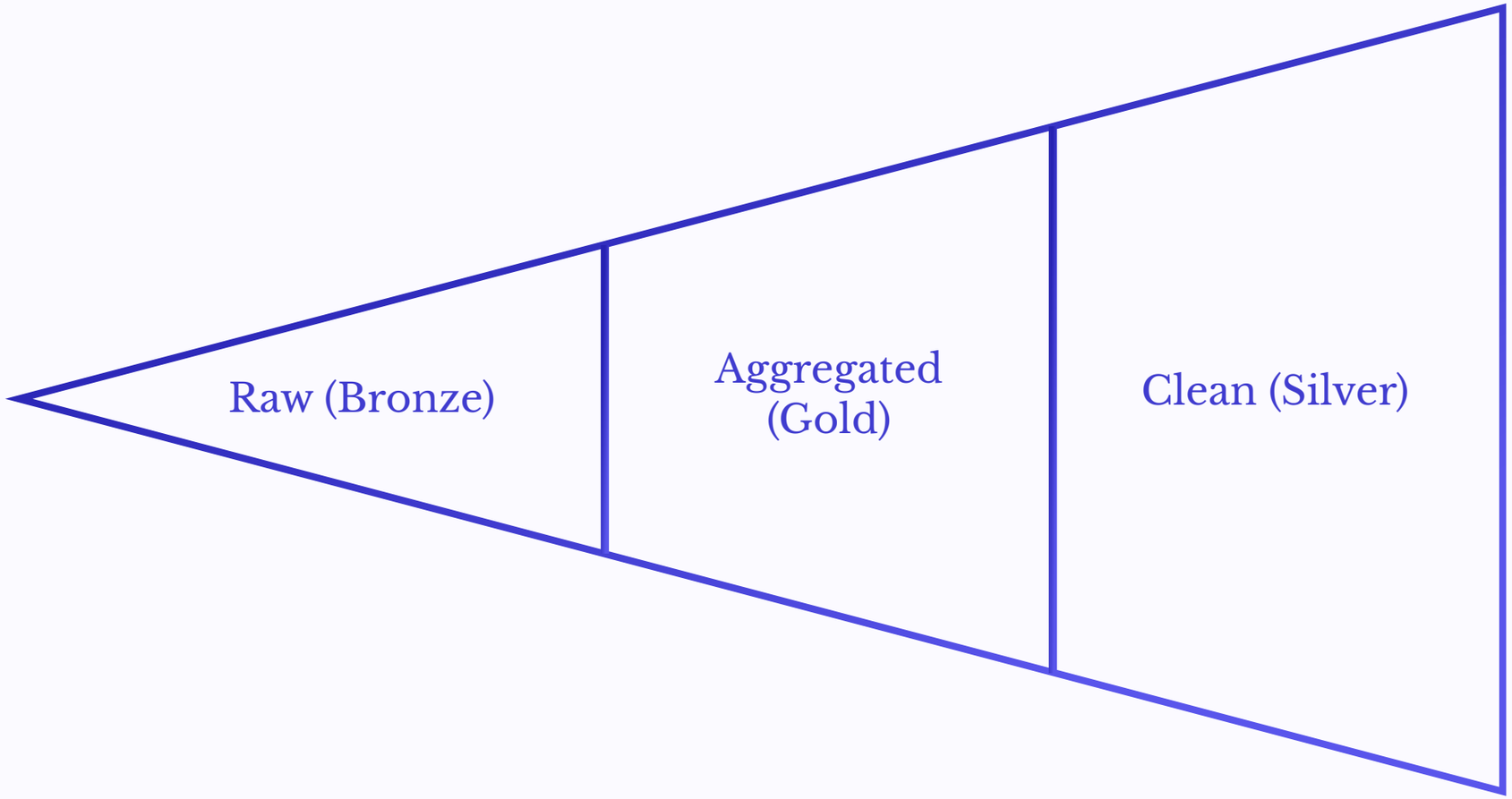
🥇 Gold

Business-ready layer. Aggregated, domain-specific datasets optimised for BI reporting, executive dashboards, and ML feature stores.

Each layer builds trust progressively. Engineers debug at Bronze, analysts work in Gold, and data scientists often blend Silver and Gold for feature engineering.

Data Flow Through the Medallion Layers

Data moves through the architecture in a well-defined path — from raw ingestion all the way to aggregated, consumption-ready outputs. Each transition adds reliability and business context.



What triggers a flow?

Pipelines can be triggered on a schedule, by file arrival events, or via streaming micro-batches — all managed through the same orchestration layer.

Idempotency matters

Each transformation should be designed to be safely re-runnable. Delta Lake's ACID guarantees make it straightforward to overwrite partitions without risking partial states.

The Role of Delta Lake

Delta Lake is the open-source storage layer that makes the Lakehouse possible. It sits on top of object storage and adds a transaction log that tracks every change to the data — turning a simple file system into a fully governed data platform.

ACID on Object Storage

Delta's transaction log serialises concurrent writes and reads, ensuring consistency even when hundreds of jobs touch the same table simultaneously.

Time Travel & Versioning

Every write creates a new version. Teams can query any prior snapshot with `VERSION AS OF` or `TIMESTAMP AS OF` — enabling rollback, auditing, and reproducible ML experiments.

Open Format

Built on Parquet with an open log protocol. Not vendor-locked — compatible with Apache Spark, Flink, Hive, Presto, and a growing ecosystem of query engines.

Why This Matters for Engineers

For data engineers and platform teams, the Lakehouse model changes the day-to-day reality of building and maintaining data infrastructure — in meaningful, practical ways.

1 Simpler Pipelines

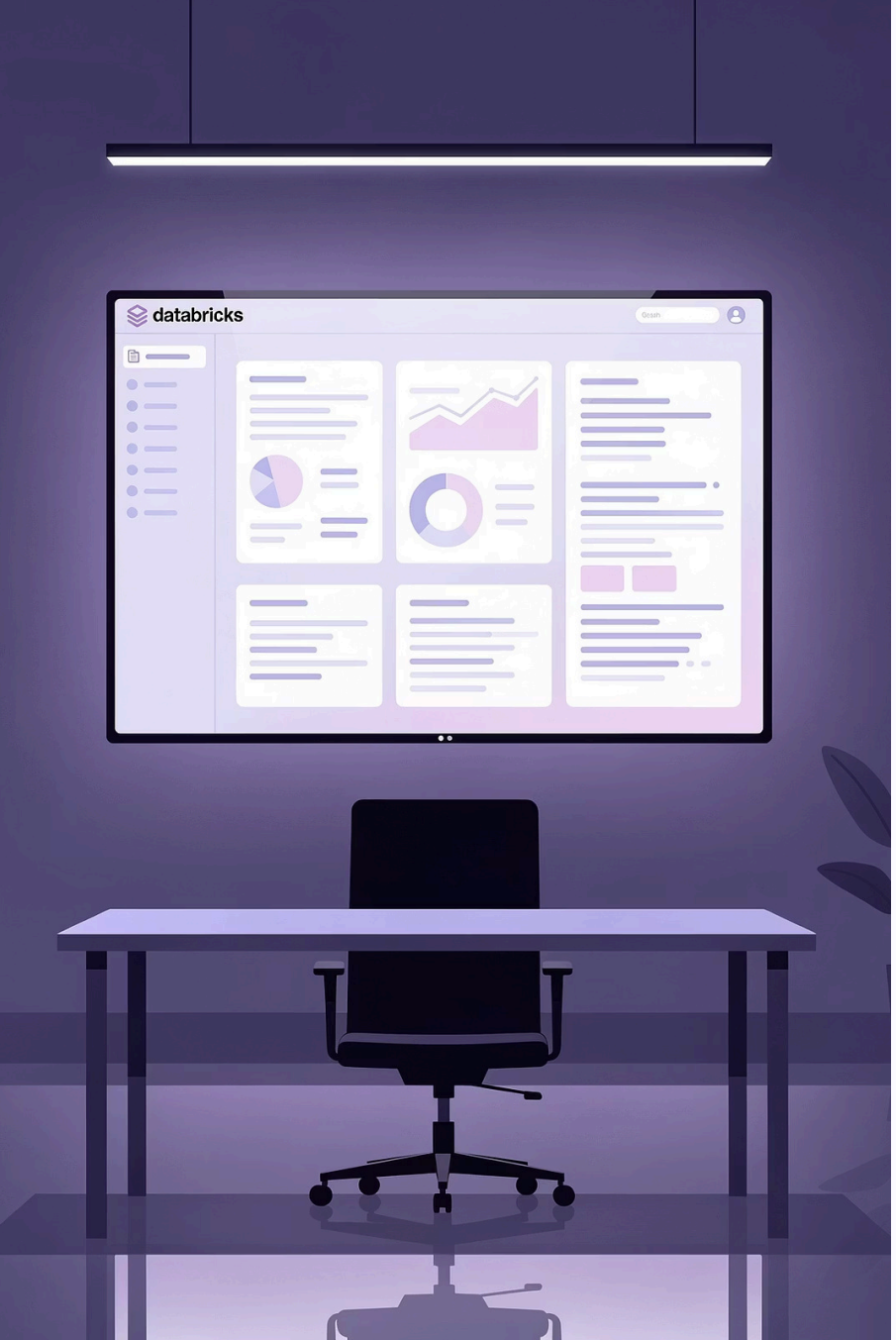
Without a separate warehouse sync layer, pipelines are shorter and easier to reason about. One ingestion path, one transformation layer, one consumption layer — governed end-to-end by Delta transactions.

2 Less Duplication & Drift

Maintaining parallel copies of data in a lake *and* a warehouse breeds inconsistency. The Lakehouse eliminates the ETL hop between systems, reducing the surface area for data quality bugs.

3 Faster Iteration

Schema evolution, time travel, and ACID rollbacks dramatically reduce the blast radius of bad deployments. Engineers can ship pipeline changes with confidence, knowing recovery is always an option.



UP NEXT

Now Let's See How Databricks Implements This

The principles we've covered — Medallion layers, Delta Lake, ACID guarantees, and unified compute — all come to life within the Databricks platform. In the next section, we'll walk through how Databricks operationalises the Lakehouse with its **Unity Catalog**, **Delta Live Tables**, and **Photon engine**.

Governance

Unity Catalog for fine-grained access control across all assets

Pipelines

Delta Live Tables for declarative, auto-scaling ETL

Performance

Photon engine for lightning-fast query execution